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MILITARY SPECIFICATION

BEARING, BALL, ANNULAR, GENERATOR AND MOTOR GENERATOR

This limited coordination military specification has been prepared by the Naval Air Systems Command, Department of the Navy, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-B-22227A. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

This specification establishes the requirements for antifriction ball bearings primarily intended for use in aircraft electrical generators and motor-generators.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-763	Steel Bars, Shapes and Forgings-Corrosion Resisting
QQ-S-766	Steel Plates, Sheets and Strips; Corrosion Resisting
QQ-W-423	Wire, Steel Corrosion-Resisting

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3110

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MILITARY

MIL-B-197	Bearings, Anti-Friction, Associated Parts and Sub-Assemblies; Packaging of.
MIL-B-1083	Balls, Bearings, Alloy Steel.
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-C-11796	Corrosion Preventive Compound, Petrolatum, Hot Application.
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-G-27617	Grease, Aircraft and Instrument, Fuel and Oxidizer Resistant
MIL-G-81322	Grease, Aircraft, General Purpose Wide Temperature Range.

STANDARDS

FEDERAL

FED-STD-66	Steel: Chemical Composition and Hardenability.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspections by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AEROSPACE MATERIAL SPECIFICATIONS

AMS 5630	Steel, Bars & Forgings, Corrosion Resistant 51440C

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI B46.1	Surface Texture
ANSI B3.15	Load Ratings and Fatigue Life for Ball Bearings

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM A380	Recommended Practice for Cleaning and Descaling Stainless Steel Parts, Equipment, and Systems
ASTM U.C. 112	Method for Estimating the Average Grain Size of Metals

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AMERICAN IRON AND STEEL INSTITUTE

AISI 305	Steel, Stainless (Austentic)
AISI 440C	Steel, Stainless (Martensitic)
AISI 1008	Steel, Ni-Cr-Mo Carbon
AISI 1010	Steel, Ni-Cr-Mo Carbon
AISI 52100	Chrome Alloy Steel

ANTI-FRICTION BEARING MANUFACTURERS ASSOCIATION

AFBMA Standard 9	Load Ratings and Fatigue Life for Ball Bearings

DEPARTMENT OF DEFENSE

ASPR	Armed Services Procurement Regulation

2.3 Proprietary Materials

KRYTOX 240 AC Grease (Manufactured by Dupont).
MOBILGREASE 28 Grease (Manufactured by Mobil).

3. REQUIREMENTS

3.1 Dash Number Designation and Procurement Documents. The individual item requirements specified herein shall be applicable in accordance with the Dash Number Designation and the procurement document. In the event of any conflict between the Dash Number Designation and the procurement document, the later shall govern.

3.2 Materials

3.2.1 New Materials vs. Reconditioned Component Parts. Bearings purchased as new items under this specification shall not contain parts that have previously been installed in any equipment and operated. There shall be no cracks or fractures of any components. Steels shall have been inspected for relative cleanliness prior to manufacturing.

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3.2.2 Rings. Rings shall be made of AISI 52100 (E52100 of FED-STD-66) or AISI 440C (QQ-S-763 or AMS 5630) steel. All rings fabricated of AISI 440C corrosion-resistant steel shall be passivated after completion of all machining or metal removing operations and prior to assembly. The metal shall be homogeneous in structure, free from cracks, pipes, seams, laminations, suface decarburization, excessive inclusion of nonmetallic impurities and other defects which may render it unsuitable for the intended purpose. On microscopic examinations, the metal shall be free from signs of overheating. Rings shall be dimensionally stable for operation from -67°F to 385°F. Rings shall withstand temperature changes and exposures with changes in diameter not exceeding 0.0001 inch per inch of diameter. The through hardness of the rings on the Rockwell "C" scale shall be C 58 to 62. The hardness values in any one lot of rings shall not vary by more than four points Rockwell "C".

3.2.3 Balls. Balls shall be made of the same material as rings. All balls fabricated of AISI 440C corrosion-resistant steel shall be passivated after completion of all machining or metal removing operations and prior to assembly. Shepard grain size shall be eight (8) or finer when examined as specified in ASTM Standard U.C. 112. Balls shall be dimensionally stable for operation from -67°F to 385°F. Balls shall withstand temperature changes and exposures with changes in diameter not exceeding 0.0001 inch per inch. The through hardness of the balls on the Rockwell "C" scale shall be in accordance with MIL-B-1083. The balls shall meet the requirements of MIL-B-1083, grade 10.

3.2.4 Retainer. Ribbon retainers shall be made of a material such as AISI 1008 or AISI 1010 low carbon steel (for AISI 52100 steel bearings) or such as AISI 305 corrosion resistant steel (for AISI 440C steel bearings) and welded or riveted.

3.2.5 Closures

3.2.5.1 Standard Seals. Seals shall be made of a materials such as nitrile (BUNA N) impervious to or compatiable with the applied lubricants and solvents and having high abrasion resistance but minimizing the effect of rubbing contact by its high degree of flexibility. Seals shall be capable of withstanding temperatures from -67°F to 250°F.

3.2.5.2 High-Temperature Seals. Seals shall be made of a material such as fluoroelastomer (Viton) impervious to or compatible with the applied lubricants and solvents and having high abrasion resistance but minimizing the effect of rubbing contact by its moderate degree of flexibility. Seals shall be capable of withstanding temperatures from -65°F to 400°F.

3.2.5.3 Seal Retaining Wires (Snap Rings). These shall be made of a commercially acceptable steel similar to the rings and balls such as QQ-W-423 or QQ-S-766.

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3.3 Surface Finish. The workmanship shall be in accordance with commercial practice used in the manufacture of high-grade electric-motor quality bearings used in aircraft and similar critical applications requiring smooth and quiet running. Raceways shall have a well-finished appearance characteristic of a honed, polished or lapped process, and free of brinnel marks, tool marks, chatter, waves, grinding or transverse scratches, pits, rust, soft spots, raised carbides or metal and other surface imperfections. Raceways shall be free of solid non-metallic inclusions, nicks and dents which exceed 0.002 inch major dimension. Raceways shall be ground or honed to a surface finish of five (5) microinches maximum in accordance with ANSI B46.1. Cylindrical mounting surfaces, lands and faces of both inner and outer rings shall have a finish which is characteristic of grinding, honing, lapping, polishing or tumbling process and shall be free of tools marks, chatter and wavings, scratches with raised metal, pits, rust or other surface imperfections visible to the normal unaided eye and shall have a surface finish of 32 microinches maximum. Balls shall have a finish characteristic of a lapping or tumbling process and be free of rust, scratches, raised metal, scants, non-metallic inclusions, flats, dents and nicks in excess of 0.001 major dimensions.

3.4 Markings. Bearings shall have permanent markings on either ring face indicating the following:

a. Heat stabilization to 385°F.

b. Country of origin.

c. Manufacturer's name or trademark.

d. Material (AISI 52100 or AISI 440C).

3.5 Bearing Series. All bearing series are deep groove bearings characterized by full raceway shoulders on both sides of both inner and outer rings without loading grooves that permit them to support both radial loads and axial (thrust) loads in either direction at low to relatively high speeds.

3.5.1 Thin, Extra Light Series. This series is designed for applications where housing diameters are restricted and it is desireable to maintain relatively large shaft diameters.

3.5.2 Standard Light Series. This series is designed for applications involving light to medium loads combined with relatively high speeds.

3.5.3 Wide Cartridge Type Series. This series is made with standard bores and outer diameters and a single row of balls, but has the same width as double row bearings of corresponding size. The extra width offers a larger support area for shaft and housing contact and added space for the pre-packed lubricant.

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3.6 Boundary Dimensions. Boundary dimensions are as specified in the Dash Number Designation. Dimensions for the shaft or housing fillet radius and broken corners shall be to commercially accepted specifications such as specified by AFBMA.

3.7 Closures

3.7.1 Double Closures Preferred. Double-sealed bearings shall be used in all applications where design considerations will allow.

3.7.2 Removeable and Replaceable. Closure and snap ring wires shall be easily removed and reinstalled with proper working tools.

3.7.3 Detachment. Closures shall be securely attached to the outer ring by removeable snap ring wires or shall have integral stainless steel inserts so as to withstand service vibration conditions or realistic (application) radial and axial loads without becoming detached.

3.7.4 Placement. Closures shall not project beyond the inner or outer ring face when subjected to pure radial loading with balls and ring grooves centered, and shall not interfere with retainer or balls under service loads, with end play taken up, or when axially preloaded.

3.7.5 Ring Distortion. Design of the attaching members shall be such that distortion of outer rings is minimized to a point that applicable tolerance limits of the bearing as specified are not exceeded.

3.8 Internal Clearance

3.8.1 Radial Internal Clearance. Radial internal clearance (radial play) shall be as specified in the Dash Number Designation.

3.8.2 Axial Internal Clearance. Axial internal clearance (axial play or end play) is seldom, if ever, a normal specification and should not be specified in place of radial internal clearance. Axial play is dependent on radial internal clearance but is also dependent on raceway radius (curvature).

3.9 Cage. The ball bearing shall have a retainer for restraining the balls within the bearing and for equally spacing them circumferentially in the raceways of the inner and outer rings. The retainer shall be a riveted or welded two-piece ribbon retainer with spherical pockets conforming to the balls but shall also allow the lubricant to be freely admitted. Retainers shall be of a ball centered design providing adequate clearance between retainer and control land, and retainer and balls. Bearings when observed with the bearing bore in the horizontal axis shall not have excessive separator drag. No bearing shall have out-of-round retainers which simultaneously contact both the inner and outer ring land areas. Retainers shall have a smooth, bright finish characteristic of a tumbling process. They shall be free of burrs and folded over material as produced by normal manufacturing operations.

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3.10 Precision (ABEC Tolerance). Bearings procured to this specification shall conform to ABEC 3 tolerances with the exception of bore and outer diameter which shall be +0.0000/-0.0002 inch.

3.11 Lubrication. Bearings requiring grease lubricant for use shall be lubricated with 30 to 40% of the void filled with the lubricant and adequately distributed within the bearing.

3.11.1 Mobil "Mobilgrease 28" (MIL-G-81322) Grease. Bearings requiring grease lubrication and having application temperatures from -65°F to 350°F shall be lubricated with Mobil "Mobilgrease 28" grease conforming to MIL-G-81322.

3.11.2 Dupont "Krytox 240 AC" (MIL-G-27617C, Type III) Grease. Bearings requiring grease lubrication and having application temperatures exceeding the maximum temperature limitation (greater than 350°F but less than 500°F) of MIL-G-81322 grease shall be lubricated with Dupont "Krytox 240 AC" grease and made of AISI 440C corrosion-resistant steel.

3.11.3 MIL-C-11796 Class 3 (Preserved). Bearings requiring open-bearing (no closures) oil flow lubrication shall be preserved with MIL-C-11796 Class 3 but subsequently cleaned and dip lubricated with MIL-L-7808 or MIL-L-23699 prior to use.

3.12 Limiting Speed. The limiting speeds given in Table 1 represent maximum application values and minimum bearing values required of bearings procured to this specification. Speed values for double-sealed grease-lubricated bearings are based on a 3200 feet per minute linear speed of the seal bore. In the interest of standardization, the lower speed values between Viton-type sealed bearings and Buna N-type sealed bearings are specified. Speed values for open (no-closure) oil flow-lubricated bearings are based on a 300,000 dN factor.

3.13 Load Ratings. The load ratings given in Table 1 represent maximum application values and minimum bearing values required of bearings procured to this specification. The static load rating (C_o) is that static radial load which corresponds to a total permanent deformation of ball and raceway at the most heavily stressed contact of 0.0001 of the ball diameter. The basic load rating (C) is that calculated, constant, radial load which a group of apparently identical bearings with a stationary outer ring can theoretically endure for a rating life of one million revolutions of the inner ring. The rating life of a group of apparently identical ball bearings is the life in millions of revolutions that 90% of the group will complete or exceed. For a single bearing, rating life also refers to the life associated with 90% reliability. Calculation of load ratings shall be in accordance with Anti-Friction Bearing Manufacturers Association (AFBMA) Standard 9 (ANSI B3.15). Parameters of bearing material, lubrication, internal clearance, temperature, type, etc. shall be in accordance with AFBMA Standard 9. Basic load ratings given in Table 1 are based on a speed of 33 1/3 RPM.

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TABLE 1

Limiting Speeds and Load Ratings

Dash No.	Sealed & Greased (RPM)	Open & Oil Flow (RPM)	C_o (lbs)	C (lbs)
2S	15,100	20,000	680	1,180
2W	15,100	20,000	680	1,180
3T	14,500	17,600	540	1,030
3S	13,700	17,600	760	1,280
3W	13,700	17,600	760	1,280
4T	12,000	15,000	850	1,620
4S	11,400	15,000	1,370	2,210
4W	11,400	15,000	1,370	2,210
5T	10,000	12,000	830	1,700
5S	10,000	12,000	1,560	2,420
5W	10,000	12,000	1,560	2,420
6T	8,300	10,000	1,210	2,280
6S	8,200	10,000	2,230	3,350
6W	8,200	10,000	2,230	3,350
7T	7,100	8,500	1,500	2,760

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DASH NUMBER DESIGNATION

BEARING, BALL, ANNULAR, GENERATOR AND MOTOR GENERATOR

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BEARING DIMENSIONS

Dash No.	Bore mm	Bore inches	O. D. mm	O. D. inches	Width mm	Width inches
2S	15	0.5906	35	1.3780	11	0.4331
2W	15	0.5906	35	1.3780		0.6250
3T	17	0.6693	35	1.3780	10	0.3937
3S	17	0.6693	40	1.5748	12	0.4724
3W	17	0.6693	40	1.5748		0.6875
4T	20	0.7874	42	1.6535	12	0.4724
4S	20	0.7874	47	1.8504	14	0.5512
4W	20	0.7874	47	1.8504		0.8125
5T	25	0.9843	47	1.8504	12	0.4724
5S	25	0.9843	52	2.0472	15	0.5906
5W	25	0.9843	52	2.0472		0.8125
6T	30	1.1811	55	2.1654	13	0.5118
6S	30	1.1811	62	2.4409	16	0.6299
6W	30	1.1811	62	2.4409		0.9375
7T	35	1.3780	62	2.4409	14	0.5512

BEARING SERIES

T	Thin extra light
S	Standard light
W	Wide cartridge

RADIAL INTERNAL CLEARANCE

F	0.0004" - 0.0007"
L	0.0007" - 0.0011"
X	0.0011" - 0.0016"

MATERIAL

	No symbol = AISI 52100
C	Corrosion resistant AISI 440C

CLOSURES

	No symbol = open bearing
B	Buna N (or nitrile) seals (2 each)
V	Viton (or fluoroelastomer) seals (2 each)

LUBRICATION

K	Krytox 240 AC
M	Mobilgrease 28
P	Preservation (MIL-C-11796 Class 3)

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requriements specified herein, unless disapproved by the Government. The Govenment reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government Test Laboratory. The contract administration office cognizant at a (contractor's) plant is responsible for the performance of Government procurement quality assurance actions (ASPR 14-203) and shall initiate a "Request for and Results of Test" (Form DD 1222) along with the sample, and a copy of the contract containing any additional controlling special requirements, to the Government Test Laboratory: The Materials Engineering Laboratory at the NAVAIR Engineering Support Office (Naval Air Rework Facility), Naval Air Station, North Island, San Diego, CA 92135 (Attn: Code 34100, Bldg. 341); designated to perform the specialized inspections reserved to the Government when imposed in the contract.

4.3 Classification of Inspection. The inspections specified herein are classified as follows:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.4 First Article Inspection. "First Article" includes preproduction models, initial production samples, test samples, first lots, pilot models and pilot lots; and "approval" involves testing and evaluating the first article for conformance with specified contract requirements before or in the initial stage of production under a contract (ASPR 1-1901). A requirement for the first article approval is designed to assure that the contractor can furnish a product that is satisfactory for its intended use and, therefore, minimizes risks for both the contractor and the Government (ASPR 1-1902). When required by the procuring activity, bearings furnished under this document shall have met the first article inspection prior to the regular production of a contract. First article inspection requirements shall consist of (i) all applicable quality conformance inspection requirements herein and (ii) special requirements as specified by the procurement document pertaining to the bearing's intended

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application. First article inspection shall only be invoked when (i) there is a significant deviation from this specification such as a new or alternate material or design, or (ii) the contractor is manufacturing the product for the first time, or (iii) documented case history warrants.

4.5 Quality Conformance Inspection. "Quality Conformance Inspection" consists of examinations and tests performed on samples from lots for the purpose of determining conformance with the specified requirements herein (DSM 4120.3). The criteria for invoking the quality conformance inspection shall be per the applicable Quality Assurance provision as determined by the Government.

4.5.1 Inspection of Product for Delivery. Inspection of product for delivery shall consist of group "A" and "B" inspections when applicable as prescribed by the Dash Number Designation. Inspection of product for delivery shall include group "C" inspection when applicable as predicated by the Dash Number Designation and when documented case history warrants as determined by the Government. Group "D" inspection is generally not performed unless justified and in such case may be more appropriately handled via First Article Inspection.

4.5.2 Inspection Lot. An inspection lot shall consist of all bearings of the same contract and Dash Number Designation.

4.5.3 Sample. The sample taken from each lot shall be random and representative in accordance with MIL-STD-105.

4.5.4 Sampling Plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for special inspection level S4. Critical, major and minor defects shall be as defined in MIL-STD-105. The acceptable quality level (AQL) shall be as follows:

a. Critical 0.65

b. Major 1.00

c. Minor 2.50

4.5.5 Disposition of Sample Units. Sample units which have passed all the applicable group inspections may be delivered on the contract, if the lot is accepted and the sample units are still within specifications.

4.5.6 Classification of Inspections. Group inspections shall consist of the inspections specified in Table 2.

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4.5.6.1 Group A. Nondestructive inspections to demonstrate product compliance with contractual requirements. Group A inspections examine characteristics most affected by variations in production processes or skills and functions vital to successful completion of the design mission.

4.5.6.2 Group B. Generally nondestructive inspections that are more complex or of a longer duration than Group A. Group B inspections examine characteristics more affected by part or equipment quality and less affected by variation in production processes and skills, functions requiring special fixtures or environments, and test that are more complex and of longer duration than Group A. Tested articles may usually be offered for acceptance with little or no refurbishment.

4.5.6.3 Group C. Periodic, generally destructive tests of characteristics depending upon product design and materials. Group C inspections are more complex tests usually including simulated service environments, are generally destructive or require major refurbishment before tested articles can be used by the services.

4.5.6.4 Group D. Destructive tests or tests of long duration that consume all or a considerable portion of design service life. Group D inspections for tested articles are considered completely expended and normally will not be accepted for delivery under contract.

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TABLE 2
GROUP INSPECTIONS

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
GROUP A		
Visual		
Surfaces	3.3	4.6.1
Markings	3.4	4.6.1
Bearing Series	3.5	4.6.1
Closures	3.7.1	4.6.1
	3.7.2	4.6.1
GROUP B		
Boundary Dimensions	3.6	4.6.2, 4.6.2.5
Ring Distortion	3.7.5	4.6.2, 4.6.2.5
Internal Clearance	3.8	4.6.2, 4.6.2.6
Cage	3.9	4.6.1
Precision	3.10	4.6.2, 4.6.2.5
Lubrication	3.11	4.6.3
GROUP C		
Materials	3.2	4.6.1, 4.6.4
Surface Finish(Metrology)	3.3	4.6.5
Closures	3.7.3	4.6.6
	3.7.4	4.6.6

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4.6 Methods of Inspection

4.6.1 Visual. Bearings shall be examined up to a maximum of 10X magnification utilizing light optics microscopy to determine conformance with the requirements of 3.2, 3.3, 3.4, 3.5, 3.7.1, 3.7.2 and 3.9. Defects or irregularities observed at up to a maximum of 10X magnification may be observed under higher magnification utilizing scanning electron microscopy as well as light optics microscopy for identification of the defect or irregularity.

4.6.2 Measurements

4.6.2.1 Standards. The standards for measurement shall be gage blocks, plug gages, ring gages, or other measuring equipment calibrated and traceable to National Bureau of Standards calibration at a temperature of 68°F (20°C).

4.6.2.2 Temperature. Measurements may be made at a room temperature other than the calibration temperature of the gages, provided that the measurements are corrected to the calibration temperature of the gages.

4.6.2.3 Repeatability. All measurements shall be made with suitable apparatus which will permit duplication of measurements to within one-half the last decimal unit specified.

4.6.2.4 Rounding Off. Measurements shall be rounded off to the accuracy indicated before judgement is made on the conformance of any item with the replacement. For example, to round off a number to four figures to the right of the decimal point (Example: 0.0001), discard all digits to the right of the fourth figure. If the discarded number is less than half a unit in the fourth place, leave the fourth digit unchanged. If the discarded number is greater than half a unit in the fourth place, add one to the fourth digit. If the discarded number is exactly half a unit, leave the fourth digit unaltered if the fourth digit is an even number, but increase it by one if it is an odd number.

4.6.2.5 Dimensions. Boundary dimensions and precision (ABEC Tolerances) shall be measured by procedures acceptable to the Anti-Friction Bearing Manufacturers Association (AFBMA) to determine conformance with the requirements of 3.6, 3.7.5 and 3.10.

4.6.2.6 Internal Clearance. Measurements shall be obtained with the bearing clean and unlubricated and with no seals. The maximum and minimum of readines taken at three different outside diameter positions (120° settings) shall constitute the radial internal clearance of a particular bearing to determine conformance with the requirements of 3.8.

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4.6.3 Lubrication. The lubrication shall be tested by utilizing light optics microscopy and infrared spectrophotometry to determine conformance with the requirements of 3.11.

4.6.4 Materials. The following are test methods to determine conformance with the requirements of 3.2.

4.6.4.1 Metallic Identification. Test methods utilizing X-ray energy spectrometry or comparable technology may be used for identification and analysis of rings, balls, retainer, and snap ring wires or any other metallic bearing components.

4.6.4.2 Non-Metallic Identification. Test methods utilizing infrared spectrophotometry or comparable technology may be used for identification and analysis of seals or any other non-metallic bearing components.

4.6.4.3 Dimensional Temperature Stability. The bearing shall be held at 375°F± 20°F for 300 hours and then measured. The bearings shall then be cold soaked at -67°F± 10°F for 300 hours and then measured. Both measurements shall be made at room temperature. Bearings that show a dimensional change of more than 0.0001 inch per inch of diameter shall be considered to have failed. Bearings subjected to dimensional stability tests shall not be delivered on the contract and shall be scrapped.

4.6.4.4 Seal Temperature Stability. Bearings with seals installed shall be sujected to temperatures of -67°F and 325°F for 100 hours at each temperature. Seals showing non retention of lubricant, deterioration, or embrittlement due to temperature shall be considered to have failed.

4.6.4.5 Passivation Test. The passivation test solution shall meet the requirements of ASTM A380, paragraph 7-C-3. The test bearings components (rings and balls) shall be immersed in the test solution, which is maintained at room temperature 70° to 75°F (21°-24°C), for six (6 ± 0.5) minutes. The bearing components shall then be gently washed with clean running water. (Clean water is defined as distilled or demineralized water with a maximum total solids content (soluable or insoluable) of 10 parts per million and a PH between 5.5 and 7.5). Each component shall be examined under 10X magnification for any evidence of copper deposit. The presence of copper deposit indicates a failure.

4.6.5 Surface Finish (Metrology). Measurements shall be made with equipment meeting the requirements of ANSI B46.1 to determine conformance with the requirements of 3.3. The equipment shall include means to provide permanent strip or polar chart type recordings. In deep groove raceways, the width of the surface is the distance from the bottom of the raceway to either land corner adjacent to the raceway.

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4.6.6 Closure Detachment and Placement. Test methods utilizing tensile or compression technology may be used for determining conformance with the requirements of 3.7.3 and 3.7.4.

5. PACKAGING

5.1 Packaging Requirements. Bearings shall be level-A packaged in accordance with MIL-B-197 method IA-8. Intimate packaging shall consist of sealed nylon bags. The following identification markings shall appear on the exterior unit packs. (The words "national stock number", "item description", "quantity" and "unit of issue" shall not be made a part of the markings.)

a. National stock number

b. Specification dash number designation and revision letter

c. Manufacturer's federal supply code for manufacturers and part number

d. Item description

e. Quantity and unit of issue

f. Contract number

g. Level of protection and manufacturer's package date

e.g., 3110-00-686-1353
M22227-2SLMB B(AS)
MFR. 21335 PM202PPFA2758
BEARING, BALL
1 EACH
DLA500-79-M-AA39
A 5/79

5.2 Packing. Bearings, packaged in accordance with 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

Custodian:

Navy - AS

Preparing Activity:

Navy - AS

Project: 3110-N586

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